NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



VRAM GUIDE

GPU VRAM Guide for Generative AI

Understand memory requirements before choosing a GPU

NexaStore Knowledge Series • 2026 Edition

1. Why VRAM Matters

VRAM is the GPU's high-speed working memory. For local generative AI, it may hold model weights, KV cache, activations and other runtime data. A powerful GPU with insufficient VRAM can be less useful for a target model than a slower GPU with enough memory to keep more of the workload on-device.

VRAM tier	Typical educational positioning	Example workloads
Integrated / shared	Cloud-first or lightweight local AI	AI-enabled applications, small supported models
8 GB	Entry local AI	Small quantized LLMs, embeddings, CV prototyping
12 GB	Mainstream AI development	Stronger local inference and moderate CV workloads
16 GB	Advanced mobile AI	More headroom for local models, context and creator/AI workloads
24 GB	High-end local AI	Larger model experimentation, advanced CV and professional workloads
32 GB+	Workstation tier	Large local experiments, higher concurrency and specialist workloads

2. VRAM Is Not the Whole GPU

Factor	Why it matters
Compute architecture	Determines supported features and performance characteristics.
Memory bandwidth	Affects how quickly the GPU can move model data.
Software ecosystem	Framework and runtime compatibility can determine whether the GPU is useful at all.
Power limit	Laptop GPUs with the same family name can perform differently at different power envelopes.
Cooling	Sustained AI workloads need thermal headroom, not only short benchmark performance.

3. Workload Planning

Workload	VRAM direction	Additional considerations
Embeddings	8 GB can be sufficient for many local experiments	Batch size, model choice and framework
Small quantized LLM	8-12 GB	Context length and quantization
14B-class local LLM	12-16+ GB	Model format, quantization and CPU offload
Computer vision	8-16+ GB	Image size, batch size and model architecture
Fine-tuning with adapters	12-24+ GB	Model size, optimizer state and training method
Large local inference	24-32+ GB	May still require offloading or multi-GPU/cloud resources

4. Laptop GPU vs Desktop GPU

Factor	Laptop GPU	Desktop GPU
Portability	Excellent	None
Power envelope	Restricted	Higher
Cooling	Limited by chassis	More thermal headroom
Upgradeability	Usually none	Often replaceable
VRAM options	Generally lower ceiling	Broader high-VRAM choices
Best fit	Mobile AI development	Sustained/high-end local AI

5. Recommendation Rules for NexaStore

• Never recommend a GPU using VRAM alone; verify software/runtime compatibility.

• Do not infer exact VRAM from a narrative guide when structured SKU data exists.

• Treat model-size guidance as approximate until model format and quantization are known.

• For laptop recommendations, include power, thermals, weight and battery trade-offs.

• If a user needs occasional heavy compute, compare the cost of premium local hardware with cloud GPU usage.

6. Example Agent Reasoning

Question	Knowledge + live data needed
'Which 16 GB VRAM laptop is in stock?'	Inventory/product tool for exact VRAM and stock; no RAG needed for the factual filter.
'Why is 16 GB better for my local LLM work?'	Retrieve this guide and the Local LLM Hardware Requirements Guide.
'Which laptop should I buy for 14B models under my budget?'	Product/price tools + RAG knowledge + recommendation reasoning.

Educational notice

This guide is part of the fictional NexaStore educational knowledge base. Hardware guidance is illustrative and model/runtime requirements should be verified for real-world use.